

Election Process Education Assistant

1. Product Overview

Build an interactive assistant that explains the election process, timelines, and step-by-step actions in a simple, trustworthy, and easy-to-follow way. The assistant should help users understand how elections work, what happens at each stage, and what they need to do at different points in the process.

Problem Statement

Many users do not understand election timelines, eligibility, registration steps, voting procedures, and post-election steps. Information is often scattered, technical, or hard to trust.

Product Goal

Create a guided assistant that makes election education clear, accessible, and interactive for first-time voters, students, and general citizens.

Success Criteria

- Users can understand the election process in under 3 minutes.
 - Users can find the correct step based on their current situation.
 - Users trust the information because it is structured, sourced, and up to date.
 - Users can complete common tasks such as checking eligibility, learning deadlines, and understanding voting steps.
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2. Target Users

Primary Users

- First-time voters
- Students and young adults
- Citizens who need a refresher on the election process

Secondary Users

- Civic educators
 - NGOs and awareness groups
 - Support staff answering election-related questions
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3. Core User Needs

- Understand the election process from start to finish
 - Know what stage the election is currently in
 - Learn the next action they should take
 - Get reminders about deadlines and important dates
 - Understand voting-day procedures
 - Clarify common terms in plain language
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4. Functional Scope

4.1 Key Features

1. **Interactive Election Timeline**

2. Shows the full election cycle in sequential steps.
3. Highlights the current stage.
4. Explains what happens next.

5. **Step-by-Step Guidance**

6. Breaks the process into simple cards or chat prompts.
7. Explains each step using beginner-friendly language.

8. **Eligibility Checker**

9. Helps users understand whether they may be eligible to vote.
10. Can ask location-specific or age-related questions where applicable.

11. **Deadline and Reminder System**

12. Registration deadlines
13. Candidate filing dates
14. Voting day reminders
15. Result announcement timelines

16. **FAQ Assistant**

17. Answers common questions such as:

- How do I register?
- What documents are needed?
- What happens at a polling station?
- What if I cannot vote in person?

18. Glossary Mode

19. Explains civic and election terms in simple words.

20. Examples: constituency, ballot, polling station, turnout, counting.

21. Location-Aware Information

22. Adapts content based on country/state/region.

23. Shows the correct rules and timeline for the user's location.

24. Accessibility Support

25. Simple language

26. Large text support

27. Screen-reader friendly UI

28. Voice input/output optional

5. User Experience Flow

Flow 1: New User Learning the Process

1. User opens assistant.
2. Assistant asks: "What would you like to know?"
3. User selects: Learn election process.
4. Assistant shows a timeline with stages.
5. User taps each stage for details.
6. Assistant offers next actions and useful links.

Flow 2: User Checking Their Status

1. User asks: "Can I vote?"
2. Assistant asks for location and age-related details if needed.
3. Assistant evaluates eligibility.
4. Assistant shows result with explanation and next steps.

Flow 3: User Needs Deadline Help

1. User asks: "When is the registration deadline?"
 2. Assistant detects location and election cycle.
 3. Assistant shows relevant deadline and explains what to do before it.
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6. Information Architecture

Main Sections

- Home / Start Here
- Election Timeline
- Eligibility
- Registration
- Voting Day
- Results
- FAQ
- Glossary
- Help / Sources

Content Hierarchy

- **Level 1:** Main stage of election process
 - **Level 2:** What happens in this stage
 - **Level 3:** What the user needs to do
 - **Level 4:** Official sources or related links
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7. Product Requirements

7.1 Must Have

- Guided election timeline
- Conversational Q&A assistant
- Step-by-step explanations
- Location-based content support
- Admin content management
- Source citations or references
- Mobile-friendly UI

7.2 Should Have

- Reminder notifications
- Voice support
- Multilingual support
- Save progress / bookmark topics

- Quick action buttons

7.3 Nice to Have

- AI-generated summaries
 - Personalized learning path
 - Chat history
 - Push alerts for election milestones
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8. Non-Functional Requirements

- Fast response time
 - Accurate and verified content
 - High availability during election periods
 - Secure handling of user data
 - Accessible for users with disabilities
 - Easy to update by content admins
 - Scalable for heavy traffic spikes
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9. Content and Data Strategy

Data Sources

- Official election commission or government sources
- Verified civic education materials
- Internal reviewed content library

Content Rules

- Use plain language
- Avoid political persuasion
- Keep information neutral and factual
- Show last updated date
- Reference official sources where possible

Content Management

- Admin panel for updating timelines, FAQs, and location-specific rules
 - Review workflow before publishing content
 - Version history for auditability
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10. Suggested System Architecture

Frontend

- Web app or mobile app
- Chat interface + guided cards
- Timeline visualization
- Search and FAQ navigation

Backend

- Python-based API layer (FastAPI recommended for performance, or Django for full-stack capabilities)
- RESTful or GraphQL APIs for frontend communication
- Business logic for eligibility, timelines, and rules engine
- Notification service (Celery + Redis for background jobs)
- Integration with CMS for dynamic content

AI Layer

- Retrieval-based assistant for answering questions from approved content
- Guardrails to prevent unsupported or biased responses
- Fallback responses when data is unavailable

Storage

- User preferences
 - Session state
 - Content library
 - Logs and analytics
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11. Recommended Tech Stack

For a Fast MVP

- **Frontend:** React / Next.js
- **Backend:** Python (FastAPI or Django REST Framework)
- **Database:** PostgreSQL
- **Cache:** Redis
- **Search:** Elasticsearch or built-in vector search
- **CMS:** Strapi / Contentful / custom admin panel
- **Hosting:** AWS / GCP / Azure

For AI Assistant Layer

- Retrieval-augmented generation over approved content
- Prompt guardrails for safety and neutrality
- Logging for quality and review

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12. API Modules

Public APIs

- Get election timeline
- Get eligibility rules
- Get deadlines
- Search FAQs
- Get glossary term explanation

Admin APIs

- Create/update content
 - Approve/review published content
 - Manage regions and elections
 - View analytics and question trends
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13. Suggested MVP Scope

MVP Features

- One region/country support
- Election timeline
- FAQ assistant
- Eligibility checker
- Glossary
- Basic admin content management
- Source-backed answers

MVP Out of Scope

- Advanced personalization
 - Multi-region complex logic
 - Full voice assistant
 - Push notifications in v1 if timeline is tight
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14. Delivery Plan

Phase 1: Discovery

- Finalize geography and election type
- Collect official sources
- Define legal/compliance boundaries
- Validate user journeys

Phase 2: Design

- Create wireframes
- Design conversational flows
- Define content structure
- Approve branding and accessibility standards

Phase 3: Build MVP

- Build frontend and backend
- Connect content database
- Add assistant logic
- Implement admin CMS

Phase 4: Testing

- Functional testing
- Content accuracy review
- Accessibility testing
- Load testing for election spikes

Phase 5: Launch

- Release pilot version
- Monitor analytics and question gaps
- Improve based on user feedback

15. Risks and Mitigations

Risk: Incorrect or outdated election information

Mitigation: Use official sources, update timestamps, admin review flow, and content versioning.

Risk: Political bias concerns

Mitigation: Keep content neutral, fact-based, and reviewed by civic content owners.

Risk: Legal or regional complexity

Mitigation: Launch with one region first and expand only after validation.

Risk: Low user trust

Mitigation: Show sources, explain logic clearly, and avoid unsupported AI answers.

16. Acceptance Criteria

The product is ready when:

- A user can understand the full election process through the assistant.
 - A user can locate deadlines and next steps based on their region.
 - Answers are accurate, neutral, and sourced.
 - The assistant works smoothly on mobile and desktop.
 - The content team can update information without engineering support.
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17. Example Assistant Prompts

- "Explain the election process step by step."
 - "What should I do before voting day?"
 - "Am I eligible to vote?"
 - "What documents do I need?"
 - "Show me the timeline for this election."
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18. Final Product Vision

The assistant should feel like a patient civic guide: clear, trustworthy, interactive, and easy for anyone to use, especially first-time voters who need simple explanations instead of legal or political jargon.